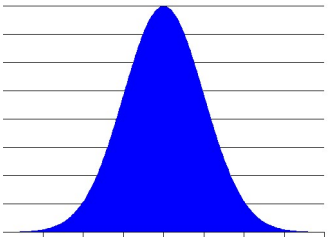
Why Statistics?

- Deal with uncertainty in repeated scientific measurements.
- Draw reliable conclusions from data.
- Design valid experiments and draw reliable conclusions.
- Be a well-informed member of society.

2

Population and statistical inferences

A population is the entire group of objects about which you want information.



Population is completely identified: Both the shape and the parameters are known.

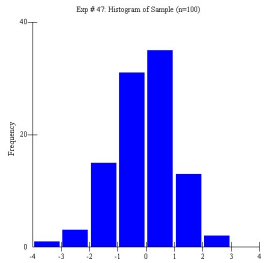
Compute probabilities, expectations, etc., for random variables X from this population.
X : future observations from this population

Deductive inferences (Tumdengelim)

3

Sample and statistical inferences

A sample is a subset of observations selected from the population. It is used to gain knowledge about the population.



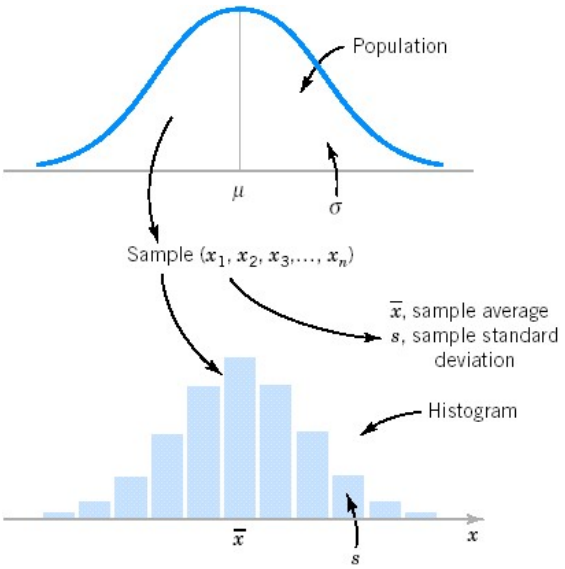
Estimate the unknown parameters of this population.

The shape of the population is known, but not its parameters.

Inductive inferences (Tumevarim)

4

Relationship between a population and a sample



5

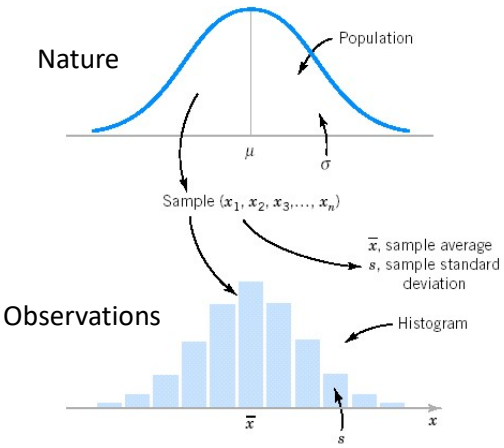
Games online?

- Have you played Tavla online?
- How do you think the dice are rolled?

- Can we also simulate a die roll? → Labs!

6

How to link the two worlds? Population and Sample



- If we can play both sides, we can understand the link between them better!
- Simulations where you can generate die rolls, coin tosses and more → representing the God who controls the Nature
- Analyzing the data that you yourself generated will show you how you can use the data to understand the population

7

In the Labs:

- You will play the God sometimes, and yourselves in other times…
- God: Simulations (many data sets can be generated)
- Yourself: Statistical analysis via Python (just one data set is there to analyze)

8

In the lectures:

- You will:
  - understand the link between the two worlds mathematically,
  - learn which questions to ask and how to answer them when you have a problem and a corresponding data set.

9